

ECE 501 Project Week 4 Report

Project Title: Vehicle Orientation Estimation Using Conventional Image Processing

Group Number: 14

Project number: 8

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Github Repository Link: https://github.com/hetpatel5107/DIP_2026_Project8_Group14

- **Introduction**

In Week 4, we will proceed further with the Sobel method as we did in week 3 for a single cropped image. This week, we will compare Sobel results we get with ground-truth results from annotations from DRASHTI-HaOBB dataset for 20 cropped images to get a better understanding of the sobel method, to check how accurate it is.

The goal of this week is not to improve the method, but to test it honestly on more than one image and to see what the result tells us about using a dominant gradient direction as a simple orientation estimate.

- **Reading the Ground-Truth Annotations**

Code:

```
with open("0bGO_UHIP_zpWU.txt", "r") as f:

    lines = f.readlines()

lines = lines[1:]

vehicle_types = []

ground_truth = []

for line in lines:

    parts = line.split()

    vehicle_types.append(parts[8])

    ground_truth.append(float(parts[10]))
```

Explanation:

The text file used in above code (0bGO_UHIP_zpWU.txt) is text file from DRASHTI-HaOBB dataset for image (0bGO_UHIP_zpWU.jpg) which has data about vehicles in image. We extract useful information from the file which is vehicle class and orientation of the vehicle.

- **Converting Ground-Truth Angles to a 0–90° Scale**

Code:

```
ground_truth_converted = []

for angle in ground_truth:

    angle = angle % 180

    if angle > 90:

        angle = 180 - angle

    ground_truth_converted.append(angle)

print(ground_truth_converted)
```

Explanation:

The dataset heading angles goes from 0° to 360°. The Sobel estimate can only lie between 0° and 90°. To compare ground-truth angle and sobel angle, ground-truth angles needs to be converted to 0°-90° scale. To do that, it is initially reduced by modulus 180° and any greater angle than 90 is brought back to 0°-90° scale. For example, $181.38^\circ \rightarrow 1.38^\circ$, $270^\circ \rightarrow 90^\circ$, and $101.4^\circ \rightarrow 78.6^\circ$.

- **Applying the Sobel Method to the 20 Vehicle Crops**

Code:

```
estimated_angles = []

for image_file in image_files:

    img = cv2.imread(image_file)

    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)

    sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

    gradient_magnitude = np.sqrt(sobel_x**2 + sobel_y**2)

    gradient_magnitude = np.uint8(

        255 * gradient_magnitude / gradient_magnitude.max()

    )

    gradient_direction = np.degrees(

        np.arctan2(sobel_y, sobel_x)

    )

    angles = gradient_direction.flatten()

    magnitudes = gradient_magnitude.flatten()

    hist, bins = np.histogram(

        angles,

        bins=180,

        range=(-90, 90),

        weights=magnitudes )

    dominant_angle = bins[np.argmax(hist)]

    orientation = abs(dominant_angle)

    estimated_angles.append(orientation)
```

Explanation:

This is the same code from Week 3, in which we got orientation using the sobel method for a single image. This week it runs on 20 cropped images without any change in parameters. For each cropped image, the color image is converted to grayscale, Sobel X and Y give horizontal and vertical intensity change, and combined into a gradient magnitude $G = \sqrt{G_x^2 + G_y^2}$ and a gradient direction $\theta = \text{atan2}(G_y, G_x)$ at every pixel. The direction of all pixels are then converted into a 180 angle histogram between -90° and 90° . After weighting bins by gradient magnitude, the highest bin in the histogram shows the most dominant gradient direction which we use to show the orientation of the vehicle.

It is important to note that Sobel only extracts local image gradients, and we make an assumption that the dominant gradient of the cropped image can be used as a simple estimate of the vehicle orientation.

- **Results: Comparison with Ground Truth**

Code:

```
results = pd.DataFrame({  
    "Vehicle": image_files,  
    "Type": vehicle_types,  
    "Ground Truth": ground_truth,  
    "Ground Truth (0-90)": ground_truth_converted,  
    "Sobel Estimated (0-90)": estimated_angles  
})  
  
results["Error"] = abs(  
    results["Sobel Estimated (0-90)"] -  
    results["Ground Truth (0-90)"]  
)
```

```

results

mae = results["Error"].mean()

minimum_error = results["Error"].min()

maximum_error = results["Error"].max()


print("Mean Absolute Error:", round(mae, 2), "degrees")

print("Minimum Error:", round(minimum_error, 2), "degrees")

print("Maximum Error:", round(maximum_error, 2), "degrees")

```

Output:

Mean Absolute Error: 33.9 degrees

Minimum Error: 1.32 degrees

Maximum Error: 67.38 degrees

	Vehicle	Type	Ground Truth	Ground Truth (0-90)	Sobel Estimated (0-90)	Error
0	v1_Sedan_17.1deg.jpg	Sedan	17.10	17.10	11.0	6.10
1	v2_SUV_181.38deg.jpg	SUV	181.38	1.38	63.0	61.62
2	v3_MotorCycle_1.68deg.jpg	MotorCycle	1.68	1.68	3.0	1.32
3	v4_MotorCycle_36.32deg.jpg	MotorCycle	36.32	36.32	5.0	31.32
4	v5_MotorCycle_183.62deg.jpg	MotorCycle	183.62	3.62	71.0	67.38
5	v6_Sedan_270.0deg.jpg	Sedan	270.00	90.00	23.0	67.00

6	v7_SUV_181.88deg.jpg	SUV	181.88	1.88	51.0	49.12
7	v8_MotorCycle_43.68deg.jpg	MotorCycle	43.68	43.68	21.0	22.68
8	v9_PickUp_178.04deg.jpg	PickUp	178.04	1.96	48.0	46.04
9	v10_MotorCycle_270.0deg.jpg	MotorCycle	270.00	90.00	79.0	11.00
10	v11_MotorCycle_270.0deg.jpg	MotorCycle	270.00	90.00	71.0	19.00
11	v12_MotorCycle_270.0deg.jpg	MotorCycle	270.00	90.00	36.0	54.00
12	v13_SUV_250.0deg.jpg	SUV	250.00	70.00	33.0	37.00
13	v14_Van_0.0deg.jpg	Van	0.00	0.00	29.0	29.00
14	v15_AutoRicksaw_79.9deg.jpg	AutoRicksaw	79.90	79.90	27.0	52.90
15	v16_Van_101.4deg.jpg	Van	101.40	78.60	51.0	27.60
16	v17_MotorCycle_77.9deg.jpg	MotorCycle	77.90	77.90	73.0	4.90
17	v18_AutoRicksaw_294.3deg.jpg	AutoRicksaw	294.30	65.70	50.0	15.70
18	v19_AutoRicksaw_0.0deg.jpg	AutoRicksaw	0.00	0.00	45.0	45.00
19	v20_Tanker_177.4deg.jpg	Tanker	177.40	2.60	32.0	29.40

Explanation:

For each vehicle, the orientation error is the absolute difference between the Sobel estimate and the converted ground-truth. Out of 20 images, code also gives output for the mean absolute error, minimum error and maximum error.

- **Error plot**

Code:

```
plt.figure(figsize=(12, 5))

plt.bar(

    range(1, 21),

    results["Error"]

)

plt.xlabel("Vehicle")

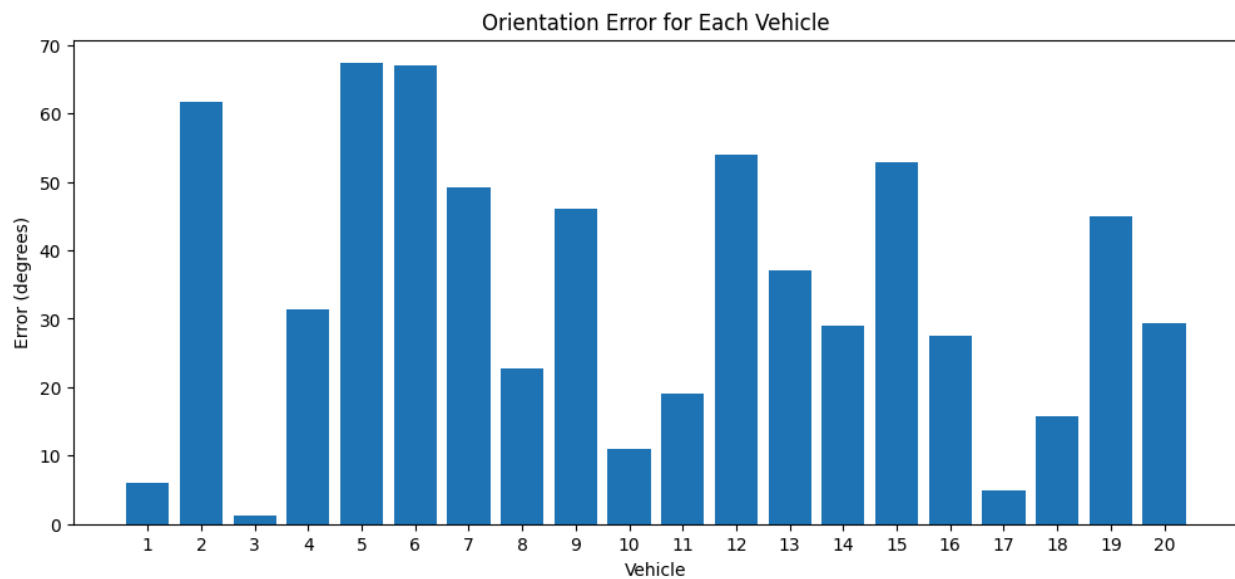
plt.ylabel("Error (degrees)")

plt.title("Orientation Error for Each Vehicle")

plt.xticks(range(1, 21))

plt.show()
```

Plot:



- **Observations**

Mean absolute error is 33.9° , minimum error is 1.32° , maximum error is 67.38° . There were some reasonable close estimates with errors like 1.32° , 4.90° and 6.10° . But there were a lot of large errors like 67.38° , 67.00° and 54.00° .

It might seem that Sobel is inaccurate, but it would not be perfect reasoning. This large error comes from the way the dominant gradient is being used as an orientation estimate. So, this method just considers the strongest gradient a factor for deciding orientation, even though these gradients could be due to vehicle boundary, roof edges, windows, windshield, internal structures, shadows, or the road lane markings.

Sobel has no idea of which edges belong to the longitudinal axis of the vehicle. The strongest edge direction is therefore not necessarily the longitudinal direction of the vehicle. As a result, the dominant gradient direction can differ significantly from the actual direction, as seen in the vehicles with errors above 45° .

- **Conclusion**

The main limitation identified is that Sobel extracts local intensity gradients without distinguishing between edges that indicate the vehicle's longitudinal direction and other edges caused by vehicle structures, shadows, or the background. Therefore, the dominant gradient direction does not necessarily correspond to the actual vehicle orientation.

To address this limitation, we will explore the Radon transform, a conventional image-processing technique that analyses the overall line structure of an image across different angles to estimate its principal orientation. Unlike simply selecting the strongest local gradient direction, the Radon transform considers the combined structure of the image, making it a promising approach for estimating the orientation of vehicles with elongated shapes.

The next stage of the project will be to implement and test the Radon-based approach on the same vehicle crops and compare its results with the existing Sobel-based baseline. This will help us determine whether analysing the overall structure of a vehicle can improve orientation estimation.

1. Work done this week

- Applied the existing Sobel-based method to 20 cropped vehicle images from the DRASHTI-HaOBB dataset.
- Obtained a dominant gradient direction for each vehicle crop.
- Converted the ground-truth angles to a 0° – 90° scale for comparison.
- Created a results table containing the ground-truth angles, estimated orientations, and orientation errors.
- Calculated the mean, minimum, and maximum orientation errors.
- Plotted the orientation error for each vehicle.
- Identified limitations of using the dominant gradient direction as a simple vehicle-orientation estimate.

2. Work to be done next week

- Research other conventional image-processing techniques for estimating vehicle orientation.
- Study how techniques that analyse the overall shape and structure of a vehicle could improve orientation estimation.
- Explore the Radon transform as a possible method for estimating the principal orientation of a vehicle.
- Test a suitable method on the existing cropped vehicle images.
- Compare its estimated orientations with the ground-truth annotations and evaluate the results.

3. Challenges faced

- Converting the dataset's 0° – 360° heading angles to a 0° – 90° scale for comparison.
- Observing large differences between the estimated orientations and ground-truth values for several vehicle crops.

- Understanding why strong gradients caused by vehicle structures, shadows, and background features can affect the estimated orientation.